Paulo Andrés Laguna Tovar

Full-Stack Software Engineer | Go · Flutter · Python

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SUMMARY

Full-stack engineer with 10+ years building software products end to end as the sole engineer of two companies. I built Flutter mobile and desktop apps, Go backends, and offline-first systems for rural areas. I migrated a full backend from Python to Go, cutting server costs by about 60%, and ran all production infrastructure.

SKILLS

Backend	Go (Echo), Python (Django, Django REST Framework), PHP (Laravel), Node.js, REST APIs
Mobile & desktop	Flutter/Dart, React Native, Electron
Frontend	Flutter/Dart, JavaScript, Astro, Angular, HTML/CSS, jQuery
Databases	PostgreSQL, SQLite, MySQL, MariaDB, Firebase (Firestore)
Infrastructure	Linux, Kubernetes, Docker, systemd, DigitalOcean, AWS, Azure, offline local networks
Hardware	fingerprint readers, signature pads, POS printers, cameras
Other languages	Java (hardware integration), C# and GDScript (game development)

EXPERIENCE

Co-founder & Lead Full-Stack Engineer — Infinity Tolima Resources S.A.S. Tolima, Colombia · 2021 – 2026

Traceability platform for gold from small-scale miners, connecting the ANMAUCOL miners' association with a precious-metals trading company. Buyers of investment bars (1, 10 and 50 g) could check each bar's full origin through a QR code.

- Sole engineer for the whole product: architecture, backend, mobile, web, servers and support.
- **GailagGold:** desktop system running the company's full operation: gold purchasing, warehouse and inventory, miner registration, and the certificate of origin required by Colombia's National Mining Agency, signed digitally by the miner. Integrated a DigitalPersona fingerprint reader, Topaz signature pad, camera and printers (letter, POS and invoice) through custom Java utilities. Fully rewrote it from Angular + Electron to Flutter for performance.
- **PriceLock:** Flutter mobile app where miners checked gold prices and posted sale offers, optimized for low-end Android phones after React Native proved too slow.
- **Infinityapp:** Flutter mobile app with real-time gold prices and a price calculator based on purity and percentage.
- **Gold bar platform:** web app to register and manage bars, with QR-based traceability for customers.
- Migrated the backend from Python (Django REST Framework) to Go, reducing server costs by ~60%.
- Scaled DigitalOcean infrastructure to demand: a Kubernetes cluster at peak usage, then a single Ubuntu server running Go under systemd.
- **Results:** 500–1,000 registered miners across several municipalities at peak, and 3,800+ fully traceable gold bars.

Founder & Developer — DBPoblacional Tolima, Colombia · 2014 – 2017

Survey software to profile victims of Colombia's armed conflict, built as a faster alternative to the slow and unstable official system.

- Built an online/offline PHP system that ran in the field on a mini server and local Wi-Fi, with no internet needed.
- Cut interview time from over 2 hours (175 questions in the official system) to about 30 minutes, with 350+ questions.
- Led field data collection using the software.

RECENT PROJECTS & FREELANCE

Nian Admin — personal project 2026

Cross-platform business management and accounting system (desktop, web and mobile) with a built-in AI assistant.

- Flutter client and Go server, with multi-tenant PostgreSQL isolated by row-level security (RLS).
- AI assistant that explains workflows, opens modules, and queries or records data through tool calls, always with user confirmation; works with several model providers and has an offline mode.
- Automated tests against real PostgreSQL (testcontainers) and Flutter UI tests.

Llantas El Toro (Chile) — freelance 2026

- Company website built with Astro, delivered with an admin manual for the client: llantaseltoro.cl

EDUCATION

Technical Degree in Systems — ITFIP 2017 – 2021
Earned within the Systems Engineering program (7 semesters completed)

International Business — Universidad del Tolima 2013 – 2017
7 semesters completed

LANGUAGES

Spanish: native · **English:** professional working proficiency (B2); translated commercial contracts and worked with international partners.